CS330 Operating Systems & Lab.

Spring 2026

Instructor : Insik Shin

Instructor

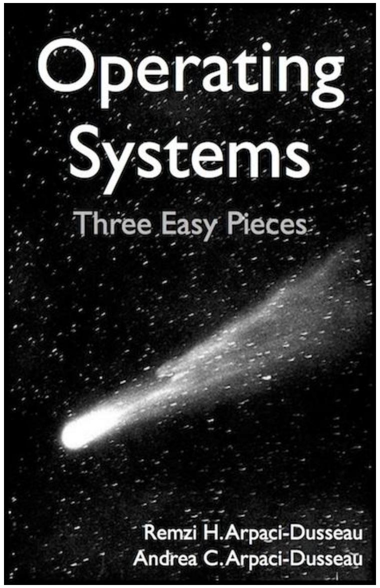
- Insik Shin
  - Professor, School of Computing
  - E3-1 Rm #4425 (CPS Lab), http://cps.kaist.ac.kr/~ishin
    - CS330
      - Spring 2009, Fall 2009 , Fall 2010, Fall 2011, Fall 2012, Fall 2013, Spring 2014, Spring 2015, Spring 2016, Spring 2017, Spring 2018, Spring 2019, Spring 2020, Fall 2021, Fall 2022, Fall 2024, Spring 2025
    - CS530
      - Fall 2009, Spring 2010, Spring 2011, Spring 2012, Spring 2013, Fall 2015, Fall 2016, Fall 2017
  - Research interests
    - Operating Systems, Mobile Computing, Systems Security
    - Mobile AI Agent

eager to explore what we can do with software, in particular, with systems + AI

Logistics

- Lecture time & place
  - Time : Mon & Wed, 13:00-14:30, Monday, 19:00-22:00
  - Place : E3-1 1501 (제1공동강의실)
  - Exam
    - Time: Monday 13:00~15:45

- Textbook
  - Operating Systems: Three Easy Pieces



- Course webpage

  http://klms.kaist.ac.kr

Prerequisites

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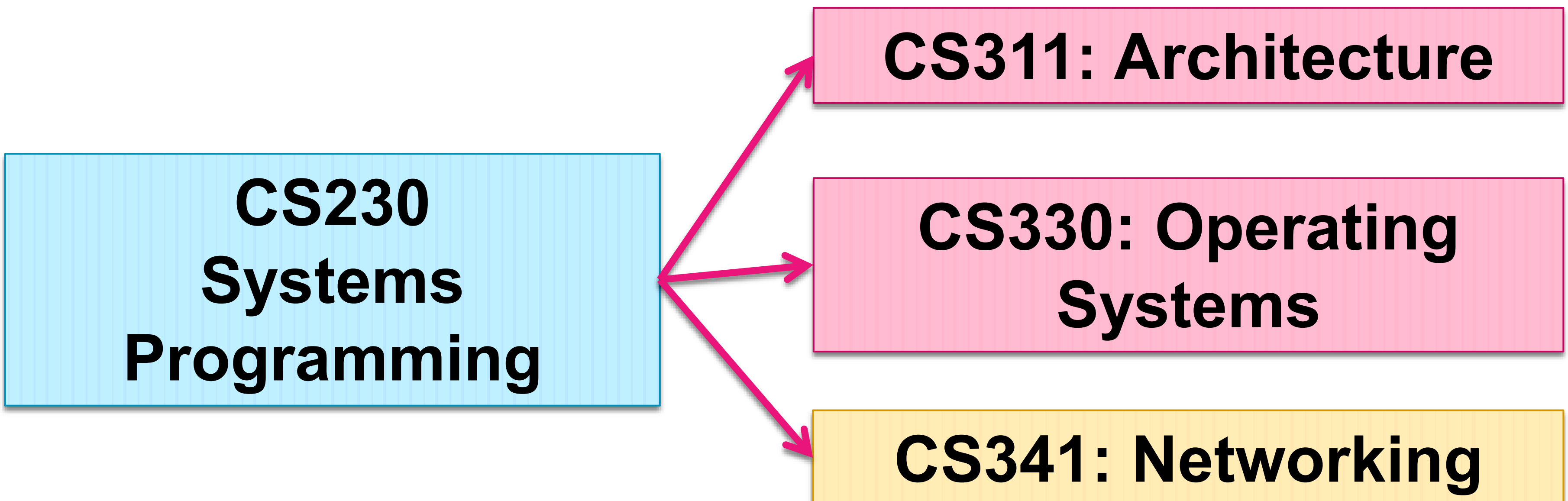
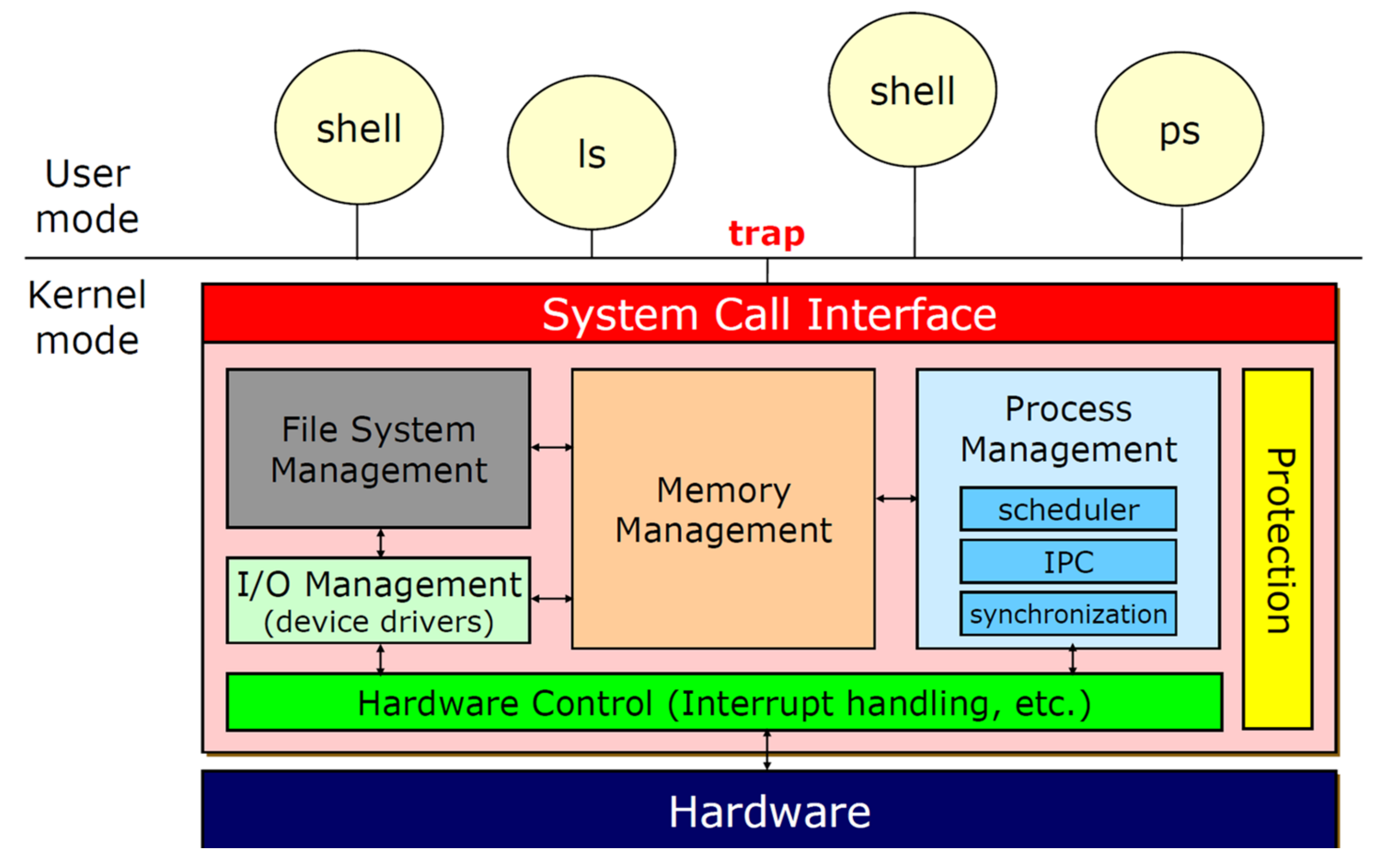
  - CS230 is strongly recommended
  - Basic understanding of key OS concepts
    - process, system call, virtual memory, file
      - the schedule of programming project goes ahead of lecture



Course Goals

- Operating systems concepts
  - Understanding OS makes you a more effective programmer



Course Goals

- Programming skills
  - Programming assignments much larger & much more difficult than many other courses
  - Many students will consider this course very hard

KeOS: KAIST educational Operating System

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- An educational project designed for…
  - Learning core operating system concepts
  - Implementing a minimal yet functional kernel f rom the ground up
- Designed with the principle of simplicity for
  - Focusing on the most essential aspects of OS development
  - Reducing debugging time for catching obscure bugs (e.g., race condition

https://github.com/casys-kaist/KeOS

Grading

- Final grade
  - 40% : Exams (Final only, no Midterm)
  - 50% : Projects (project quizzes & test scores)
  - 10% : Homework, quiz & class participation

- Plagiarism policy
  - We will be very serious on it
  - Cite any code that inspired your implementation
  - You should develop your own logic / code

Schedule

Week	Lecture	Project	Week	Lecture	Project
1	Intro.		9	Virtual memory	3
2	OS Structures	1	10	Virtual memory	3
3	Process	1	11	File systems	3
4	Threads	2	12	File systems	4
5	CPU scheduling	2	13	Mass storages	4
6	Synchronization	2	14	I/O systems	4
7	Deadlocks	3	15	Mobile systems	
8	Midterm Exam		16	Final Exam	